

Telegraphy is made of two words: tele meaning at a distance and graphy meaning to write. With this, long-distance communication of text messages was conceptualised. In fact, this was an improved version of a semaphore.

Discovery of radio communication paved the way for developing electrical telegraphy. Using semaphore networks, early telegraphy was carried out by French scientist Claude Chappe.

The credit for commercial telegraphy goes to Samuel Morse. A code known as Morse code was developed around 1837 for signalling of alphabets. Accordingly, text messages, as a series of codes, were transmitted between the transmitter and receiver using electrical signals.

Telegraph as the first long-haul communication system got systemised in postal communication. Post offices at a distance served as transmitters and receivers. But telegraphy, unlike natural voice communication, did not have features such as interactive, instantaneous, fully wireless, man-mobile, unique addressed and multi-mode communication. It connected only two geographical points at a time for service.

Electrical communication that started its journey with the telegraph is still marching ahead in multiple advanced forms.

Long-distance voice communication developed over the years

The next long-distance electrical communication is known as the telephone. The word telephone is made up of two words: tele meaning at a distance and phone implying sound. In a telephone, interactive voice communication is made over a long distance in real time (zero latency). In this system, telephone handsets act as transmitters/receivers (transceivers). These connect geographical points for the purpose of communication without any support for mobility.

All handsets have a microphone and an earphone. The microphone converts voice signal into electrical signal, and acts as the transmitter. Whereas, the earphone acts as the receiver. Electrical signals get transmitted over either wired or wireless mode, between two phones connected at a distance.

Alexander Graham Bell invented

the telephone in 1876. The first call he made was to his assistant on March 10, 1876. His first words over his invention were, "Mr Watson, come here. I want to see you."

The telephone operates via unique identification numbers that change from one location to another. Like a telegraph, the telephone connects geographical locations without the support of man-mobility. But unlike telegraph, it is interactive and supports real-time communication. It is therefore a step ahead of telegraph, but behind natural communication, as it does not support features such as multi-mode communication, fully wireless link, man-mobility and unique identification irrespective of location, among others.

Mobile communication—commonly known as cell telephony—was developed to infuse some sort of mobility in voice communication. To achieve this, geographic location-attached and portable handsets were replaced.

In mobile communication, portable handsets function using wireless (radio) links. Cellular or cell communication uses portable cellphones. The phones are registered in a geographic area called cell. Like post offices, each of which have a geographical area of operation, cells are connected to cover wider geographical areas for communication.

This achieved the target of mobility. But this is primarily machine mobility, and not man-mobility of natural voice communication. Unless you carry the cellphone with you, mobile communication is not possible.

Several global standards are used for cellular communication. A few of these are GSM (Global Standard/Systems for Mobile), IS 95, 200 (international standard) and UMTS (Universal Mobile Telecommunication Systems).

With advancements in information technology such as integrated circuit (IC) technology, cell communication is now equipped with transportation of text messages, data, video along with interactive voice communication over a distance. Long-distance, interactive and real-time voice, video, text and data (hence, multimedia) communication over wireless links is now possible with mobile communication.

However, challenges of unique identification of communicating enti-

ties and man-mobility for natural voice communication remain unresolved. With international roaming, of course, the issue of unique identification has been addressed to some extent.

Moving beyond cellphones, advancement of the Internet led to the support of multimedia real-time communication in many forms. The Internet is basically a data-carrying network. With the advent of VoIP (Voice over Internet Protocol), mobile IP and IPv6 (Internet Protocol version 6) technologies, the Internet can now carry multimedia over long distances, in no time.

The challenge of unique identification has been resolved further by the Internet. As an example, an e-mail ID can be used by only one unique person for communication. But, man-mobility still remains unresolved.

Personal communication network (PCN) was conceived, thereafter, to resolve the issue of man-mobility and unique identification for long-distance multimedia, interactive and long-haul communication. The idea first came about in the UK as a tool for a very-high-speed digital wireless cellular communication network with an intelligent system. Phones on the Move is a variation of personal communication, which is undergoing research in the UK. Development of PCN technology will pave the way for technology-enabled natural communication over long distance.

Data communication and networking

Stephen Hawking said, "We are all now connected by the Internet, like neurons in a giant brain." This begs the question, why were we not connected by any of the predecessors of the Internet, such as telephony or telegraphy? The answer is, the Internet is a network that has converted BAD IT (where B stands for bursty, A for asymmetric, D for delicate and IT for insensitive to time) into an excellent network for all purposes.

In communication, all purposes cater to three basic services: voice, video and data. The predecessor of the Internet, namely, telephony, aimed to support voice and video communication. Telephony systems and networks cannot carry data. Characteristics of data are different from those of voice and video.